The acronym UART stands for “Universal Asynchronous Receiver Transmitter” – it is name given to a Peripheral which supports Serial Communication using a certain protocol.

This name was suggested by a committee called as Electronics Industry Association(EIA) & sometimes also called as Electronics Industry Alliance(EIA).

UART peripheral can be offchip peripheral also in the form of an IC example Intel’s 8250,8251 & 16550 etc. Off Chip UART chips/ICs are peferrably used with a microprocessor since these don’t come which On Chip UART peripheral.

Can also be used with microcontrollers in case in the need for additional Off Chip UART peripheral apart from On Chip UART peripheral for any typical application of Embedded System’s Requirement.

OR

else a microcontroller with more than one On Chip UART peripheral can be selected for a typical application of Embedded System’s Requirement.

OR

A Soft UART (i.e Bit Banging approach of UART Peripheral’s Protocol) Could also be possibly Implemented through the use of General Purpose I/O pins and software statement instead of dedicated hardware peripheral/logic for same typical requirement.

In the Intel’s 8051 architecture & compatible versions of microcontroller(s) there is ONLY provision for ONE ON CHIP UART Peripheral. But in Extended Versions/Variant of 8051 family of controllers there may be provision for 2 or more ON CHIP UARTs.

Features:
- UART supports serial, wired, full duplex communication.
- 2 data wires(TxD,RxD) available in uart communication
- It supports asynchronous communication.
- If data is transmitted along with the clock then it supports synchronous communication. (USART)
- It is for shorter distance communication

Advantage:
- Simple hardware and s/w

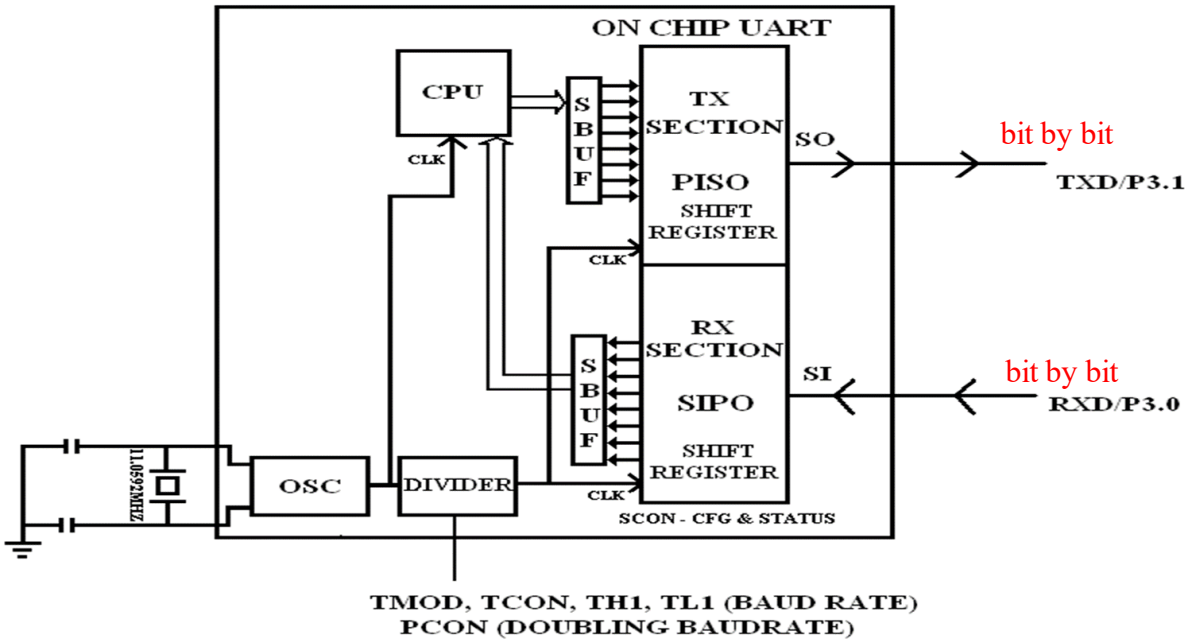
Disadvantage:
- No Handshaking signl
- No much error checking methods

Some peripherals which are working on UART communication:
- GSM Module (Global system for mobile communication)
- GPS Receiver (Global positioning system)
- RFID Reader (Radio frequency Identification)
- Smart Card reader/writer
- Finger print module
- Bluetooth modules
- Xbee modules
- Resistive Touch screen module
- UART based bar code scanner ..... etc

Now considering ONLY Intel’s 8051 architecture & compatible versions of microcontroller the SFRs associated with ON CHIP UART peripheral are name SCON, SBUF, TMOD, TCON, TH1, TL1, PCON for Without Interrupt Based for UART Communication.
Addition IE & IP for Interrupt Based UART Communication

SBUF is physically two registers.
One is write only and is used to hold data to be transmitted out of the 8051 via TxD.
The other is read only and holds received data from external sources via RxD.
Both mutually exclusive registers use address 99h.

Generic Diagram of On-Chip UART Peripheral:



UART Frame Format:

Start bit : always logic 0
Data bits : 5 to 9 bits
Parity bit : 0 or 1
Stop bit : always logic 1

1 start bit	5 to 9 Data Bits	0 to 1 parity bits	1 to 2 stop bits

------ data frame -------

---------- UART FRAME FORMAT ----------

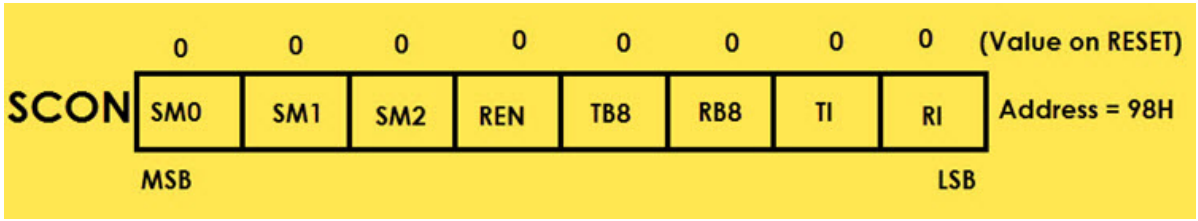
standard 8051 UART frame format: start bit | 8-data bits |stop bit

NOTE: UART supports lower significant bit (LSB) of data transmission/Reception first.

SCON (Serial Port Control Register):
Is Used for Configuring the Mode(s) of Operation of the UART Peripheral, as Synchronous Serial Communication Mode Or Asychronous Serial Communication Modes.

The On Chip UART supports 1 Mode of Sychronous Serial Communication & 3 Modes of Asychronous Serial Communication – namely mode0(synchronous), mode1,mode2,mode3(asychronous).

Additionally SCON register can also be used for monitoring the status of Serial Data transmision/Reception Completion



SCON - bit as well as byte addressable SFR

SM0	SM1	Mode	Description
0	0	0	8-bit UART;baud =f/12
0	1	1	10-bit UART;baud =variable
1	0	2	11-bit UART; baud =f/32or f/64
1	1	3	11-bit UART;baud =variable

SM2: used for Multiprocessor communication
REN: set/cleared by software to enable/disable reception
TI: Transmit interrupt flag. Set by HW at the begin of the stop bit in mode1 and cleared by software
RI: Receive interrupt flag. Set by HW at the begin of the stop bit in mode1 and cleared by software

By default tranmitter is enabled, so no need to set any bit for transmitting the data from microcontroller to external device

UART MODE1:
10 bits are transmitted through TXD or received through RXD

a start bit (0) , 8 data bits (LSB first) and a stop bit (1)

On receive, the stop bit goes into RB8 in SCON

Baud rate is variable

BAUD=((2^SMODxCF)/(12x32x(256-TH1)))

If SMOD=0 in PCON it IMPLIES
=1 x (CF/(12x32x(256-TH1)))

TH1 = 256 - ((CF / 384) / BAUD)

if CF= 12 MHZ, and baud rate 9600

TH1 = 256 - ((12x10^6)/384)/9600
= 256 - ((12000000/384)/9600)
= 256 - (31250/9600)
= 256 - 3.255
= 252.745

if CF= 11.0592 MHZ, and baud rate 9600

TH1 = 256 - ((11.0592x10^6)/384)/9600
= 256 - ((11059200/384)/9600)
= 256 - (28800/9600)
= 256 - 3
= 253

8051 UART tolerence level is 0%.

transmit:
SBUF=data;
wait for TI flag
receive:
wait for RI flag
read the SBUF data